

NIYOGUSHIMWA Natanael
Musanze - Rwanda
nathanaelniyogushimwa@gmail.com
0781796824
02/02/2025

Rwanda Water Resources Board
Nyarugenge Pension Plaza, KN 3 Avenue
P.O Box: 6213 Kigali, Rwanda
Email: hackathon@rwb.rw

Subject: Motivation Letter for Water Resources Modelling Hackathon 2025

Dear Selection Committee,

I am writing to express my keen interest in participating in the Water Resources Modelling Hackathon 2025, organized by the Rwanda Water Resources Board. As a third-year undergraduate student pursuing an Advanced Diploma in Information Technology at Rwanda Polytechnic, Musanze College, I am highly motivated to contribute to this hackathon with my technical expertise and passion for sustainable solutions.

My academic journey has been deeply rooted in leveraging technology to address real-world challenges. With a strong background in software engineering, data science, and programming, I have developed skills in Python, machine learning, and geospatial analysis—tools that are essential for water allocation modeling and erosion control. My prior research on machine learning algorithms to enhance network security has equipped me with a problem-solving mindset that I believe will be valuable in analyzing and simulating water availability and erosion control mechanisms.

Participating in this hackathon aligns perfectly with my career aspirations of using technology to drive environmental sustainability. I am particularly excited about the opportunity to apply hydrological models like HEC-HMS, QSWAT, or WEAP to analyze water distribution for sustainable resource management in the Mbirurume sub-catchment. Additionally, my proficiency in Python and R for data analysis will enable me to contribute effectively to satellite imagery processing and geospatial mapping of radical terraces for erosion control.

Beyond my technical skills, I have experience in collaborative and fast-paced problem-solving environments. My leadership role in organizing a technology conference on campus has honed my ability to work in teams and think critically under pressure—qualities that I am confident will contribute positively to the hackathon.

I am eager to bring my expertise, dedication, and innovative mindset to this competition. I look forward to the opportunity to collaborate with like-minded individuals and contribute to the development of solutions that promote sustainable water resource management and biodiversity conservation in Rwanda.

Thank you for considering my application. I would be honored to participate in this initiative and look forward to your positive response.

Sincerely,
NIYOGUSHIMWA Natanael